Anticipated questions

Preparation for 25 September 2026 — not a handout

The theory

Is the decay law new? Surely low-SNR expansions are classical.

No, and we say so in the paper. The information law is not ours: fourth-order statistics entering the Gaussian-channel expansion at second order is classical; Katsevich and Bandeira give the order-k likelihood expansion; Balanov, Bendory and Edidin give the layered SNR^k Fisher geometry, which under this channel is the same statement in a different clock; Aarts and co-authors write the forward-process cumulant attenuation. What we add is the statement at the level of the *score* with its constant $\frac{1}{6}\|\kappa_4\|_F^2$, the crossover as an exact scale, the hypothesis under which the expansion is a theorem together with the heavy-tailed regime outside it, verification against an exact non-Gaussian dependent score, and the learning consequences. If he pushes here, concede the law immediately — the contribution is the constant, the boundary of validity, and the exactness.

What exactly does your theorem require?

An exponential moment. This is a real restriction and it was tightened during the audit: without it the e^{-8t} term is not the leading one and the expansion is not a theorem. Student-t needs $\nu \geq 9$ for the eighth moments; the family ladder shows a genuine departure for heavier tails, and we report it as predicted by the moment condition rather than as a failure of the law.

Proposition 1 — is that not just LMMSE with extra steps?

It is an identification: Gaussian message closure on the chain computes exactly the LMMSE score of the covariance twin. That is what turns 0.168 from "the best we managed" into a proven floor for every second-order method, which is what lets us attribute the margin.

The comparison — where a referee will press

Are the networks just under-trained? Is this a strawman?

The honest defences, in order. The architectures are transformers and convolutional window models, screened and re-tuned *separately inside every regime*, not once at the frozen point. The best of them is reported per cell, not the average. We also report an easier interface in which the network is additionally given the *clean* sequence, which bounds what the restriction costs. And the conservatism runs against us: our own estimator is still improving when its iteration ladder runs out in 56 of 96 cells, so our reported error there is an upper bound. The weakest point to concede: we cannot prove no architecture would do better. The claim is about these architectures at this budget.

Why one noisy view per sequence? That is not how diffusion models are trained.

Correct, and we measure three interfaces, not one. The standard recipe — clean data, a fresh corruption every step — is interface A. Interface C, one noisy view and nothing else, is the honest analogue of "all you have is noisy data". It is the setting where the structural knowledge should matter most, and it is the one we claim. Do not let the claim slide into A.

You handed yourself the true model class. Is the comparison not circular?

This is the strongest objection and it has two answers, both experimental. The capacity control: within the same model class, restricting to Gaussian innovations puts us exactly on the second-order floor, so the margin is fourth-order structure rather than "knowing the family". The misspecification sweep: we perturb the data away from the assumed model and measure how far it can go before the advantage is gone — a second order of memory at 0.3–0.4, a correlation varying along the chain at 0.1–0.14. We are claiming an advantage over a measurable range, not in general.

How do you decide when to stop, without the true score?

On held-out one-view evidence, which uses no clean data and no true score. We also report what the oracle rule would have given, so the cost of being honest is visible rather than hidden: it is small, and the two rules pick the same iteration in 52 of 96 cells.

Scale, and what it means

$L = 32$ to 128, M up to 2048. Why so small?

Because the exact score requires the grid message passing, and its cost is what buys exactness. This is the trade the whole paper makes: a small, solvable laboratory in which every claim is measured against truth, rather

than a large setting in which nothing can be checked. Say it as a choice, not as a limitation we hope nobody notices.

Does a better score actually give better samples?

Yes, measured. Driven through the same reverse diffusion on identical Brownian paths, so the score is the only difference, the fitted kernel produces innovation excess kurtosis 1.680 against the reference sample's 1.94 — and close to what the *exact* score reaches under the same sampler (1.863), so most of the residue is the discretisation of the reverse process rather than our estimate. The covariance-matched score gives −0.001: an essentially Gaussian chain, the fourth-order structure gone entirely.

What does this say about real diffusion models?

Keep this narrow. It says which structure the noise destroys and in what order, that the ordering is exponential with a computable crossover, and that in a case where we can check, knowing the data's factorisation is worth a large constant factor at equal information. It does not say a structure-aware method beats a transformer on images. Do not let the conversation drift there.

Status, and integrity of the record

What is still running, and what would change?

One experiment: a sample-complexity measurement at sixteen seeds and five noise levels. Nothing in the brief depends on it. Say plainly what it is for: the corollary predicts how the number of sequences needed grows with noise level, and the earlier run *cannot* support that claim — its intervals include zero and we do not report an exponent. The new run decides whether an exponent can be claimed at all. If its intervals also include zero, that is the result.

Has anything you previously reported turned out to be wrong?

Yes, twice, and both are on the record with the cause. Our first screen of the regime comparison was void: a defect in our own code re-created the random generator inside a loop, so the training set was one sequence repeated. We found it, wrote a test that now fails on that pattern anywhere in the repository, and re-ran everything. That same test then found the identical defect in the archival capacity sweep — which is why the earlier capacity conclusion is void, and why the stopping-rule explanation previously offered for it is withdrawn. Volunteer this rather than wait for it: the pattern he should see is that the process catches its own errors.

How do I know the numbers in this document are right?

Every experiment is frozen with hash-pinned outputs and a configuration registered before it ran. Each reported number is generated from those outputs by one tool and recomputed independently by a second that fails if they disagree; the figures check themselves against the manuscript's numbers and refuse to be written on a mismatch. The paper is not compiled by hand from remembered values; nothing in it can drift from the experiment that produced it.